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SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME:Artificial Intelligence **COURSE CODE: CSA1715**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 30 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

*I _**k.yamini(192572181)**_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

k.yamini

Annexure A

Descriptive Written Test

1.A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2.A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Q1 (10)	Q2 (10)	Total Marks (20)
Understanding of Concepts	25% (2.5)	/2.5	/2.5	/5
Explanation	25% (2.5)	/2.5	/2.5	/5
Application	20% (2)	/2	/2	/4
Organization	15% (1.5)	/1.5	/1.5	/3
Accuracy	10% (1)	/1	/1	/2
Clarity	5% (0.5)	/0.5	/0.5	/1
Total per Question	**10 Marks**	**/10**	**/10**	**/20**

1. Smart Home Automation System

1. Introduction

A smart home automation system uses intelligent agents to manage lighting, temperature, and security based on real-time inputs such as motion, temperature, and time, along with learned user preferences.

2. Types of Intelligent Agents

a) Simple Reflex Agent

- Works on condition–action rules.
- Example: If motion is detected → turn ON light.
- Does not consider past or future states.

b) Model-Based Agent

- Maintains an internal state of the environment.
- Example: Remembers room occupancy and adjusts lighting accordingly.
- Suitable for partially observable environments.

c) Goal-Based Agent

- Acts to achieve specific goals.
- Example: Maintain room temperature at 24°C.
- Selects actions based on goal achievement.

d) Utility-Based Agent

- Maximizes performance using a utility function.
- Example: Balances comfort and energy efficiency.
- Chooses the best action among alternatives.

3. Application in Smart Home

Lighting:

- Simple reflex → ON/OFF based on motion detection
- Model-based → remembers user patterns

Temperature:

- Goal-based → maintains desired temperature
- Utility-based → optimizes comfort vs energy usage

Security:

- Model-based → detects unusual activity
- Utility-based → prioritizes safety alerts

4. Analysis and Justification

- Simple reflex agents are fast but limited.
- Model-based agents improve decisions using memory.
- Goal-based agents ensure task completion.
- Utility-based agents provide optimal performance.

Best Approach:
A hybrid agent (Model-Based + Goal-Based + Utility-Based) is most effective because it:

- Learns user behavior
- Maintains comfort
- Optimizes energy consumption
- Ensures security

5. Conclusion
A hybrid intelligent agent provides the best solution for smart home systems by combining learning, goal achievement, and optimization.

2. Robot in Maze – Uninformed Search

1. Introduction
A robot placed in an unknown maze uses uninformed search strategies to find a path to the exit without prior knowledge of the environment.

2. Uninformed Search Strategies

a) Uniform Cost Search (UCS)

- Expands the node with the lowest path cost.
- Guarantees an optimal solution.
- Uses a priority queue.

b) Iterative Deepening Search (IDS)

- Combines Depth-First Search and Breadth-First Search.
- Explores the search tree level by level.
- Uses less memory.

3. Advantages and Disadvantages

Uniform Cost Search (UCS)
Advantages:

- Finds optimal path
- Works with varying costs

Disadvantages:

- High time complexity
- Requires more memory

Iterative Deepening Search (IDS)
Advantages:

- Low memory usage
- Complete search

Disadvantages:

- Repeats nodes
- Slower compared to UCS

4. Agent Types in Maze Problem

Simple Reflex Agent:

- Cannot solve effectively due to lack of memory

Model-Based Agent:

- Stores visited paths and avoids loops

Goal-Based Agent:

- Focuses on reaching the exit

Utility-Based Agent:

- Selects the shortest or least-cost path

5. Best Combination

Recommended Approach:
Model-Based + Goal-Based Agent with Uniform Cost Search (UCS)

Justification:

- Model-based → remembers explored paths
- Goal-based → aims to reach the exit
- UCS → ensures shortest and optimal path

6. Conclusion

Combining intelligent agent design with UCS provides an efficient and optimal solution for maze navigation.